

SurveiLens: Natural Language-Based Moment Retrieval for Surveillance Video Analysis

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Abstract—Closed Circuit Television (CCTV) systems generate vast amounts of video footage daily, yet the process of manually reviewing these recordings remains highly inefficient. Security personnel often spend hours, sifting through long duration videos to locate relevant incidents, a process that delays crucial decision making and the delivery of justice. This work proposes an automated solution that eliminates the need for manual video sifting by enabling natural language-based video retrieval. Our approach leverages a sophisticated processing pipeline consisting of video frame sampling, image captioning, semantic similarity computation, and temporal localization to extract precise timestamps or video segments corresponding to a given natural language query. This system allows investigators to describe an event and rapidly retrieve the relevant portion of CCTV footage from hours of recordings. By significantly reducing the time and human effort required to analyze surveillance videos, our method can accelerate investigations and facilitate timely delivery of justice.

Index Terms—Automated Surveillance Analysis, Natural Language Video Retrieval, Temporal Event Localization, Semantic Video Search

I. INTRODUCTION

The proliferation of Closed-Circuit Television (CCTV) systems has led to an exponential growth in video data, making them integral to modern surveillance and law enforcement. This footage is often the sole witness to criminal activities, holding critical evidentiary value. However, the process of analyzing this vast repository of video remains a formidable challenge. Security and investigative personnel are frequently compelled to dedicate significant resources to manually sift through recordings to locate specific events, a task that has been identified as a key barrier to effective CCTV use [1]. This labor-intensive work not only diverts valuable human resources from higher-level analytical tasks but also introduces significant delays in critical decision-making and, consequently, the delivery of justice.

The inefficiency of manual review is particularly acute in real-world forensic scenarios. As a result, crucial evidence may remain buried within an overwhelming volume of unanalyzed data, hindering investigative progress [2]. The need for an automated, efficient, and accessible solution to this problem is therefore not just a matter of convenience, but a critical step toward a more responsive justice system.

Recent advancements in computer vision and natural language processing have spurred research into automated video analysis, particularly in the area of video moment localization using natural language queries [3]. Significant progress has been made in developing models that can identify specific moments in videos based on textual descriptions. Seminal works have explored guided attention mechanisms [4], the power of large-scale pre-trained models like CLIP for end-to-end retrieval [5], and novel video sampling and reasoning strategies [6]. Further research has focused on sophisticated cross-modal interaction networks [7], [8] and causal intervention methods to improve retrieval accuracy [9]. More recent work has explored long-form video understanding through the lens of large language models (LLMs) and agent-based systems. Approaches like **VideoAgent** [10] and **LongVLM** [11] demonstrate impressive zero-shot question-answering capabilities by decomposing long videos and using LLMs for interactive reasoning. While these methods push the boundaries of general long-form video understanding, they are primarily evaluated on VQA benchmarks. This highlights a critical research gap: existing video moment localization methods are not readily scalable for long-duration videos, and the newest long-form approaches do not address the specific task of efficient moment retrieval for practical applications like surveillance.

Table I summarizes the average video lengths (in seconds) of several widely used benchmark datasets, along with their corresponding citations. These statistics are referenced from the paper titled “Towards a Complete Benchmark on Video Moment Localization” [12].

Prompt : wood burning stove with flames

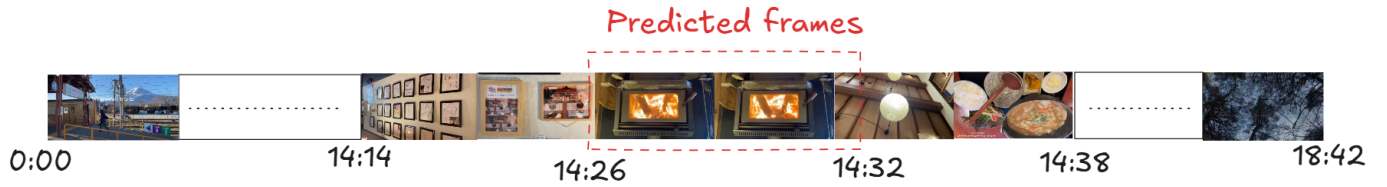


Fig. 1. Predicted keyframes from an 18-minute, 42-second video using the natural language prompt: "wood burning stove with flames".

TABLE I

AVERAGE VIDEO LENGTHS OF COMMONLY USED BENCHMARK DATASETS

Dataset	Average Video Length (seconds)	Citation
ActivityNet	117.6	[13]
Charades-STA	30.8	[14]
TACoS Multi-Level Corpus	287.1	[15]
DiDeMo	53.8	[16]
YouCook2	315.4	[17]
MSR-VTT	426.1	[18]
TVR	76.1	[19]

In this work, we address this gap by proposing a framework specifically designed for natural language based event retrieval in long form CCTV videos. Our approach is tailored to the unique constraints of forensic analysis, aiming to bridge the gap between current academic research and real world investigative needs. By enabling users to query hours of footage with simple descriptive sentences, our system drastically reduces manual effort and accelerates the evidence discovery process. The primary contributions of this paper are threefold:

- We propose a novel framework for temporal event localization that is explicitly designed to operate efficiently on long duration surveillance videos.
- We introduce a curated dataset of long form videos (20-30 minutes in length) with corresponding natural language queries and timestamp annotations to facilitate research in this domain.
- We demonstrate the practical utility of our approach, showcasing its ability to significantly reduce the time required to locate relevant events in CCTV recordings compared to manual review.

II. PROPOSED METHOD

This section details the methodologies developed and evaluated for natural language based event retrieval in long form video. Our approach is fundamentally centered on addressing the challenges posed by lengthy surveillance footage, where manual inspection is computationally and temporally prohibitive. We present a comparative analysis of various techniques within a modular pipeline, focusing on two core components: video frame sampling and the generation of semantic representations via embedding or captioning models.

A. Common Pipeline

The foundational workflow of our proposed system is designed for efficiency and scalability. Given that long form videos can contain tens of thousands of frames with significant visual redundancy, processing every frame is impractical. Therefore, the first critical step is frame sampling, which aims to distill the video into a concise yet representative set of key frames. Once this subset is extracted, we generate a semantic representation for each frame using an advanced vision-language model. A user provided natural language query is converted into a similar representation, and a similarity search is performed to identify and return the timestamps of the most relevant frames.

B. Frame Sampling Methods

The initial stage of our pipeline is dedicated to frame sampling, a critical step for managing the high volume of data in long form videos and eliminating redundant content. We evaluated the following strategies.

1) *Uniform Sampling*: Uniform sampling serves as a baseline for our comparative analysis. In this approach, frames are extracted at a fixed temporal interval, irrespective of the video's content. A significant portion of existing research on video moment localization has relied on this simple method, often sampling every n -th frame. For our experiments, we also use a baseline of sampling one frame per second. While computationally simple, this method does not account for the video's visual content, meaning it may oversample static scenes or miss brief, critical events. Our work differentiates itself by exploring more sophisticated, content-aware sampling techniques to achieve more efficient and accurate retrieval.

Algorithm 1 Uniform Frame Sampling

```

1: Input: Video frames  $V$ , frames per second  $fps$ 
2: Output: Selected Frames  $F_S$ 
3: Initialization:  $F_S \leftarrow \{\}$ ,  $n \leftarrow \text{round}(fps)$ 
4: for each frame  $I_i$  in  $V$  at index  $i$  do
5:   if  $i \pmod n == 0$  then
6:     Add  $I_i$  to  $F_S$ 
7:   end if
8: end for

```

2) *SSIM-based Sampling*: To introduce content awareness into our sampling, we employ a method based on the Structural Similarity Index (SSIM). This technique is designed to select frames that are structurally dissimilar from one another, ensuring that only frames representing a significant visual change are retained. This approach is adapted from the work on key frame identification by Roy Chowdhury et al. [20]. The SSIM metric is a measure that quantifies the perceived change in structural information between two images. For two frames, x and y , the SSIM score is calculated using the standard single-window formula, which is implemented in our chosen PyTorch library. The formula is:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (1)$$

Here, μ_x and μ_y are the local means, σ_x^2 and σ_y^2 are the local variances, and σ_{xy} is the local covariance of frames x and y . The constants $C_1 = (k_1 \cdot \text{data_range})^2$ and $C_2 = (k_2 \cdot \text{data_range})^2$ are small positive values used to prevent instability. The local statistics are computed via convolution with a smoothing kernel.

In our implementation, we use a standardized PyTorch library ('ignite.metrics.ssim') to compute the SSIM score. The process involves iterating through the video and calculating the SSIM score between the current frame and the previously selected frame. A new frame is sampled only if this score falls below an empirically determined threshold of 0.6.

Algorithm 2 Frame Structural Selection

```

1: Output: Selected Frames  $F_S = \{I_k\}$ 
2: Input: Video frames:  $\{I_l\}$ ,  $F_S = \emptyset$ 
3: Initialization:  $I_{ref} \leftarrow I_0$ ,  $step \leftarrow 2$ ,  $s \leftarrow 1$ ,  $j \leftarrow 0$ ,  $F_S \leftarrow \{I_0\}$ 
4: while  $j < l$  do
5:   while  $s > \alpha$  do
6:      $j \leftarrow j + step$ 
7:      $s \leftarrow \text{ssim}(I_{ref}, I_j)$ 
8:   end while
9:    $I_{ref} \leftarrow I_j$ 
10:   $F_S \leftarrow F_S \cup I_{ref}$ 
11: end while

```

3) *Semantic Scene Detection*: This method segments the video into coherent scenes (or shots) based on significant visual transitions. We employ a shot boundary detection tool, such as PySceneDetect, which analyzes the HSV color space of each frame to detect changes in color distribution. When the difference between consecutive frames exceeds a predefined threshold (τ_{scene}), a new scene boundary is registered. This process results in the division of the video into a sequence of semantically consistent shots, each annotated with start and end timestamps. From each detected shot, a single representative frame is selected from its temporal midpoint, ensuring a concise yet contextually rich summary of the video content.

Algorithm 3 Shot Segmentation-based Sampling

```

1: Input: Video  $V$ , threshold  $\tau$ 
2: Output: Selected Frames  $F_S$ 
3: Initialization:
4: Detect scene boundaries using ContentDetector with threshold  $\tau$ .
5: Obtain scenes  $S = \{(s_i, e_i)\}$ , where  $s_i, e_i$  are start/end frame indices.
6:  $F_S \leftarrow \{\}$ .
7: Procedure:
8: for each scene  $(s_i, e_i) \in S$  do
9:    $mid \leftarrow s_i + (e_i - s_i)/2$ 
10:   $I_{rep} \leftarrow I_{mid}$ 
11:  Compute CLIP embedding of  $I_{rep}$  and BLIP caption.
12:  Add  $(mid, \text{embedding}, \text{caption})$  to  $F_S$ .
13: end for
14: Optional Clustering:
15: if  $|F_S| > M$  then
16:   Apply MiniBatchKMeans with  $K = \min(\text{NUM\_CLUSTERS}, |F_S|)$ .
17:   Update  $F_S$  with cluster IDs.
18: end if

```

C. Embedding and Caption Models

After the frame sampling stage, we generate rich semantic representations for the selected frames to enable text-based retrieval. We explored two distinct approaches:

1) *CLIP: Text-Image Embedding*: Our first approach utilizes the Contrastive Language-Image Pre-training (CLIP) model [5] to create a shared multi-modal embedding space. CLIP is a powerful foundation model trained on a massive dataset of over 400 million image-text pairs. This extensive training allows it to learn robust, high-quality vector representations for both images and text, mapping them into the same latent space.

This shared embedding space is crucial for our pipeline, as it makes the comparison between visual and textual content highly efficient. Let the vector embedding of a sampled frame be $F \in \mathbb{R}^d$ and the embedding of the user's natural language query be $P \in \mathbb{R}^d$, where d is the dimensionality of the shared space. Since both F and P reside in the same space, we can directly compare them using a similarity metric. We compute the cosine similarity between the query embedding and each frame embedding to retrieve the top- k most relevant frames and their corresponding timestamps. The cosine similarity is given by:

$$\text{similarity}(F, P) = \frac{F \cdot P}{\|F\| \|P\|}$$

This approach enables fast and effective zero-shot retrieval without needing to fine-tune the model for our specific task.

2) *BLIP: Image Captioning for Text Retrieval*: Our second approach leverages the Bootstrapping Language-Image Pre-training (BLIP) model [21] to first generate a descriptive textual caption for each sampled frame. This process allows

us to translate the visual content of each key frame into a comprehensive textual description.

Unlike the approach with CLIP, once the descriptive captions are generated, we shift the similarity comparison from a multi-modal (image-to-text) space to a uni-modal (text-to-text) space. Both the generated captions and the user’s original query are then converted into dense vector embeddings using a pre-trained sentence transformer model. In our implementation, we use the ‘SentenceTransformer(‘all-MiniLM-L6-v2’)’ model for this purpose. This model maps sentences into a high-dimensional vector space where the distance between vectors corresponds to their semantic similarity. We use cosine similarity to identify the top- k most semantically similar captions and return the timestamps of their corresponding frames.

III. RESULTS AND DISCUSSION

In this section, we present the empirical results of our comparative study. We evaluate the different pipeline configurations on a dataset of long-form videos, focusing on retrieval accuracy and computational efficiency. Our goal is to identify the trade offs inherent in each approach and determine their suitability for practical applications.

A. Experimental Setup

Dataset: To effectively develop and validate our framework, we curated a novel dataset of 12 continuous, unedited videos, all sourced from YouTube. This was specifically designed to address the limitations of existing benchmarks, which predominantly consist of short, pre-segmented clips. The videos in our dataset range in duration from 18 minutes 52 seconds to 35 minutes 58 seconds, with an average length of approximately 24 minutes and 47 seconds. The videos’ duration distribution is highly balanced across time spans:

- **Videos < 20 minutes (33.3%):** 4 videos fall into this category.
- **Videos 20–25 minutes (41.7%):** 5 videos fall into this category.
- **Videos > 25 minutes (25.0%):** 3 videos fall into this category.

Our curation was guided by four core principles to mirror real-world surveillance challenges:

- **Long-Duration and Continuous Recording:** The videos were chosen for their extended length, which is crucial for testing the “needle-in-a-haystack” problem inherent in analyzing hours of footage. This substantial temporal space is represented by content like virtual walking tours and travel vlogs.
- **Diversity of Scenes and Environments:** To ensure generalizability, the dataset includes a wide range of settings, such as urban and rural landscapes, indoor and outdoor activities, and varied perspectives (e.g., first-person POV, static, or from a moving vehicle). This allowed us to include a large number of diverse situations and scenarios.

- **High Density of Discrete, Actionable Events:** The videos are rich with a high density of distinct, describable events. Our dataset contains a total of **55 event captions** across all videos, with a **mean of 3.67 captions per video**. This density of non-overlapping events is critical for rigorously testing the temporal localization accuracy of a retrieval model, as it requires the system to identify multiple, unique moments within a single long video.
- **Quasi-Static Camera and Naturalistic Content:** The footage often mimics fixed surveillance or body cam perspectives, capturing unscripted, everyday occurrences rather than staged actions. This makes our dataset a more suitable proxy for genuine surveillance footage.

Metrics: We evaluate our approach using two primary metrics:

- **Recall@ k (R@ k):** The percentage of queries for which the correct video segment is retrieved within the top- k results. We report R@1, R@5, and R@10.
- **Average Processing Time:** The total wall-clock time (in seconds) required for the complete pipeline - from frame sampling to returning the final timestamps.

Sampling Strategy Setup: Our experimental framework revolves around three distinct frame sampling strategies, each designed to balance efficiency and representativeness:

- **Uniform Sampling:** As our baseline, this approach has no specific parameters. It operates by simply sampling every n -th frame, where n is the integer value of the video’s frames per second (fps).
- **SSIM-based Sampling:** We used the `ignite.metrics.ssim` library in PyTorch for this method. The sampling is based on a threshold of **0.6**; a new frame is selected only when its SSIM score with the previous frame falls below this value.
- **Shot Segmentation:** For content-driven segmentation, we use the `PySceneDetect` library’s `ContentDetector` with a detection threshold of 30.0. The representative frame for each detected shot is chosen as the middle frame of its `[start, end]` span. If no shots are detected, the entire video is treated as a single scene.

Model and Preprocessing Setup:

- **BLIP Model for Captioning:** We employ the `Salesforce/blip-image-captioning-large` checkpoint for image-to-text generation. Frames are converted from OpenCV’s BGR to RGB format, then to a PIL Image before being processed by the `BlipProcessor` for tokenization, resizing, and normalization. Inference uses `model.generate()` with default (greedy) decoding and `skip_special_tokens=True`.
- **CLIP Model for Embedding:** For image-text embedding, we use the `openai/clip-vit-base-patch32` model. Images are preprocessed through the `CLIPProcessor`, and queries are tokenized similarly. Both image and text embeddings are L2-normalized for consistent cosine

similarity computation. Batch sizes were set to 64 for Uniform and SSIM-based sampling, and 16 for Shot Segmentation.

Optional Experimental Components: We further explored additional enhancements, applied *only to the Shot Segmentation approach*, to assess their effect on retrieval quality. These include:

- **Prompt Expansion:** Up to 10 paraphrases of each query were generated using the Pegasus paraphraser (tuner007/pegasus_paraphrase) and English–French–English backtranslation.
- **Semantic Filtering:** Generated paraphrases were filtered using the SentenceTransformer model (all-MiniLM-L6-v2), retaining only those with a similarity score ≥ 0.65 to the original query and uniqueness threshold ≥ 0.8 .
- **Clustering:** Shot-level CLIP embeddings were optionally clustered using MiniBatchKMeans with parameters NUM_CLUSTERS=8, batch_size=64, and n_init=10. Clustering was applied only when at least two shots were detected.

All Shot Segmentation results reported in this paper correspond to experiments conducted with these optional components enabled. We did not apply these enhancements to the Uniform or SSIM-based sampling approaches.

B. Performance Analysis

The quantitative performance of our evaluated pipelines is summarized in Table II. The results highlight significant differences in both retrieval accuracy and processing speed across the methodologies, which are visualized in the subsequent figures.

TABLE II
COMPARATIVE PERFORMANCE OF DIFFERENT RETRIEVAL PIPELINES.

Method	R@1	R@5	R@10	Time (s)	Avg. Frames
SSIM + CLIP	0.482	0.750	0.804	392.03	11,344
SSIM + BLIP	0.533	0.533	0.667	1876.37	11,344
Uniform + BLIP	0.321	0.500	0.518	339.91	1,443
Shot Segmentation + CLIP	0.393	0.625	0.661	287.51	156

Analyzing long-form video content, such as surveillance footage, presents a significant computational challenge. Given that a single video in our dataset averages 24 minutes and 47 seconds in length, a brute-force, frame-by-frame analysis is impractical. Frame sampling is therefore not just a performance optimization but a critical necessity for making such analysis feasible. This section details the dramatic reduction in frames achieved through our sampling strategies and the resulting impact on processing speed and overall efficiency.

1) *Frame Reduction and Computational Load:* The primary goal of frame sampling is to intelligently reduce the number of frames that need to be processed by computationally intensive vision-language models. Our methods achieve this

with remarkable effectiveness, cutting down the workload by up to 99%.

An average video in our dataset contains over 44,000 frames. Processing each of these frames would be incredibly time-consuming. Our three sampling strategies significantly mitigate this issue, as shown in Figure 2:

- **Shot Segmentation:** This is our most aggressive and effective strategy. It reduces the number of frames from over 44,000 to an average of just **156**. This represents a staggering 99.6% reduction, effectively isolating the most visually distinct moments and discarding redundant information.
- **Uniform Sampling:** While simpler, this method is still highly effective. It brings the frame count down to an average of 1,487, a reduction of 96.6%.
- **SSIM-based Sampling:** This content-aware method retains more frames to capture subtle visual changes, resulting in an average of 11,344 frames. While less aggressive than the others, it still achieves a 74.1% reduction in frames, providing a balance between detail and efficiency.

By processing only a small fraction of the original frames, we significantly reduce the computational demand on our models, leading directly to substantial gains in speed.

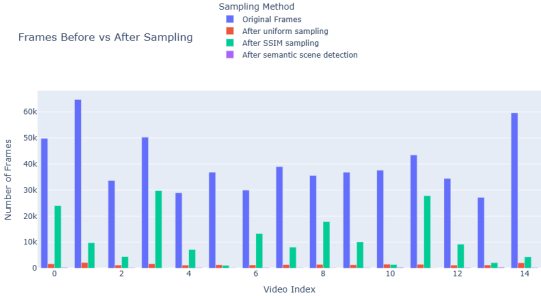


Fig. 2. Frames Before vs. After Sampling: A comparison of the number of frames across our dataset before and after applying various sampling methods.

2) Performance Boost and End-to-End Processing Time:

This reduction in frames translates directly into a massive boost in processing speed. The “total time” metric reflects the complete, end-to-end duration of our pipeline - from the initial video preprocessing and frame sampling all the way to generating the final predictions with the CLIP or BLIP models.

Contextualizing these times against the average video length of 1487 seconds (24 minutes, 47 seconds) highlights the efficiency of our approach:

- **Shot Segmentation + CLIP (Fastest):** This pipeline completes its analysis in an average of just **287.51 seconds** (about 4.8 minutes). This is more than 5 times faster than real-time playback, making it an exceptionally efficient method for quickly retrieving events from long videos.
- **Uniform + BLIP:** This approach takes **339.91 seconds** (about 5.7 minutes), demonstrating that even a simple sampling strategy can yield significant speedups.

- **SSIM + CLIP:** Balancing performance and accuracy, this method requires **392.03 seconds** (about 6.5 minutes).
- **SSIM + BLIP (Slowest):** The combination of a less aggressive sampling method and the more complex BLIP model results in the longest processing time at **1876.37 seconds** (about 31.3 minutes). Notably, this is longer than the actual video duration, underscoring the critical impact that both the sampling strategy and model choice have on overall performance.

In summary, intelligent frame sampling is the cornerstone of our efficient video analysis framework. By drastically reducing the number of frames, we enable our models to process long-duration videos significantly faster than real-time, turning an otherwise infeasible computational task into a practical and effective solution for event retrieval.

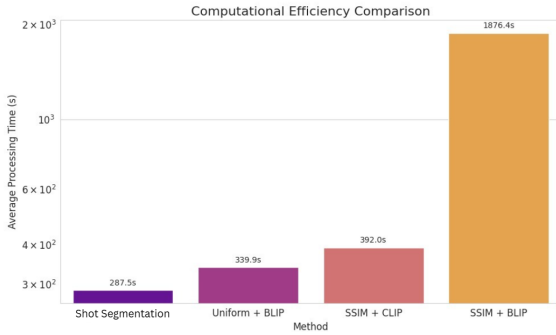


Fig. 3. Computational efficiency comparison, showing the average processing time in seconds for each method. Note the logarithmic scale on the y-axis, used to visualize the large differences in time.

3) *Analysis of Retrieval Accuracy:* In evaluating the performance of a video retrieval system, **Recall@k (R@k)** is a crucial metric. It measures the percentage of times the correct video segment is found within the top k search results. This directly reflects the model’s practical utility: a high R@1 indicates the system is precise enough to find the correct answer immediately, while high R@5 and R@10 scores show its effectiveness at bringing relevant clips to the forefront for user review.

Our experiments, as summarized in Table II and Figure 4, reveal a clear trade-off between pinpoint accuracy and broader retrieval effectiveness across the different pipeline configurations. The performance highlights distinct strengths, catering to different operational needs:

- **Highest Precision (R@1): SSIM + BLIP** For tasks demanding the highest precision from the very first result, the SSIM + BLIP pipeline proved superior, achieving the top R@1 score of 0.533. This means that for 53.3% of all queries, this method successfully identified the correct video segment as its number one result.
- **Best Overall Recall (R@5 & R@10): SSIM + CLIP** When considering a slightly broader search, the SSIM + CLIP method demonstrated the most robust performance, securing the highest scores for both R@5 (0.750) and

R@10 (0.804). This pipeline successfully found the correct event within the top 5 results for 75% of queries, and within the top 10 for over 80% of queries. This makes it the most effective “all-around” performer, providing a fantastic balance of high recall and reasonable processing speed.

- **Fastest Performance: Shot Segmentation + CLIP** The Shot Segmentation + CLIP pipeline stands out for its incredible efficiency. While not leading in recall, its scores are highly competitive and remarkable given its speed. It achieved an R@1 of 0.393, R@5 of 0.625, and R@10 of 0.661. Despite processing the fewest frames (only 156 on average), it still finds the correct clip in the top 10 results for over 66% of queries, and does so in under 5 minutes. This makes it the undisputed choice for applications requiring rapid, high-throughput analysis where near-instantaneous results are critical.

The choice of pipeline presents a clear strategic decision. For maximum precision, SSIM + BLIP is the leader. For the best balance of comprehensive retrieval and speed, SSIM + CLIP is the strongest. For scenarios where speed is the absolute priority, Shot Segmentation + CLIP offers respectable accuracy at a fraction of the computational cost.

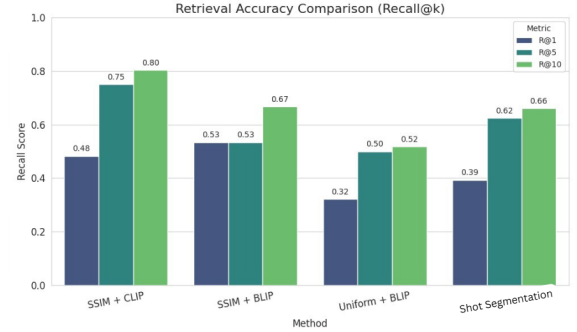


Fig. 4. Retrieval accuracy comparison of the four pipeline methods across Recall@1, Recall@5, and Recall@10 metrics.

C. Discussion

Our findings demonstrate a clear trade-off between retrieval accuracy and processing speed. The choice of both the frame sampling strategy and the vision-language model critically influences the final performance. The intensive, content-aware sampling of SSIM appears to provide a more comprehensive set of candidate frames, which benefits retrieval accuracy but incurs a significant computational cost.

Furthermore, it is important to note that our approach is specifically optimized for long-duration videos where a brute-force frame-by-frame analysis is infeasible. This method may not be as beneficial for short videos (e.g., a few seconds to a minute), where the overhead of the sampling and retrieval pipeline might outweigh the cost of simply processing all frames.

In summary, this work establishes an important performance baseline for event retrieval in long-form video. The results

suggest that different pipeline configurations are suited for different operational needs, and a deeper exploration of these complex interactions is a promising avenue for our future research.

IV. CONCLUSION

This paper addressed the critical challenge of inefficient manual review of long-duration surveillance footage. We proposed and systematically evaluated several automated pipelines for natural language-based video retrieval, combining various frame sampling techniques (Uniform, SSIM, and Semantic Scene Detection) with state of the art vision language models (CLIP and BLIP). Our empirical results establish crucial performance baselines, revealing a significant trade off between retrieval accuracy and computational efficiency. The findings demonstrate that content aware sampling methods yield higher accuracy at the cost of processing time, while different embedding and captioning models show distinct advantages for either top rank precision or broader recall.

The primary implication of this work is that the optimal retrieval strategy is highly dependent on the specific application. For forensic investigations where finding the exact moment is paramount, slower but more thorough methods like our SSIM + BLIP pipeline are justified. For rapid screening or near real time analysis where speed is the priority, faster pipelines utilizing aggressive semantic sampling are more suitable. Beyond security, this technology has wide ranging applications in other domains, such as the content creation industry for efficiently searching through raw footage, in sports analytics for tactical analysis.

For future work, our next step is to explore more sophisticated approaches to enhance both the model's capabilities and its real-world applicability. A significant avenue for improvement lies in **incorporating temporal models** to move beyond frame-level analysis. Integrating Video Transformer networks (e.g., TimeFormer, ViViT) or temporal action localization (TAL) models could allow our system to analyze sequences of frames, enabling it to understand and retrieve complex, multi-step actions as they unfold over time.

Additionally, we plan to address the domain gap through **domain-specific fine-tuning**. The performance of our vision-language models (CLIP and BLIP) could be significantly improved by fine-tuning them on a curated dataset of domain-specific footage (e.g., low-resolution CCTV video). This would adapt the models to the unique visual characteristics of surveillance data, improving their robustness to real-world conditions.

Finally, we propose developing an **interactive and feedback-driven system**. Instead of relying on a single query, future iterations could incorporate a **human-in-the-loop** approach, where the system presents initial results and refines its search based on user feedback.

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